
Hitting Time Guarantees for Blind Ballistic “Billiard” Search

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Definitions

Let the state space be a 2-dimensional discrete torus $S = \mathbb{Z}_N \times \mathbb{Z}_M$. The total size of the state space is $|S| = NM$. The agent is tasked with finding an arbitrary target node $x^* \in S$. The values of N and M are strictly unknown to the agent.

Assume without loss of generality that $N \leq M$. We enforce a bounded aspect ratio assumption, meaning there exists a constant $R \geq 1$ such that $M \leq R \cdot N$. Let the maximum dimension be $D = M$. Consequently, $D^2 \leq R \cdot |S|$, meaning $D = \Theta(\sqrt{|S|})$.

The Iterative Ballistic Process $\mathcal{B}$

Since D is not known in advance, sampling uniformly up to D is impossible. The search instead scales dynamically, operating in progressively larger *phases*.

Initialize a phase counter $j = 1$. During Phase j , the agent executes the following routine:

- (i) The agent guesses a maximum dimension $D_j = 2^j$.
- (ii) The agent runs a continuous sequence of “epochs” (linear streaks). In each epoch:
 - (1) The agent selects a direction vector v uniformly at random from the set of axial and diagonal generators: $V = \{(1, 0), (0, 1), (1, 1), (1, -1)\}$.
 - (2) The agent selects a streak length L uniformly at random from the integer interval $[1, D_j]$.
 - (3) The agent traverses the grid by taking L consecutive 1-cell steps along v , checking every node along the path for the target state x^* .
- (iii) The agent continues initiating epochs until the cumulative number of single-cell steps taken in Phase j reaches or exceeds $c \cdot D_j^2$. To preserve the translation invariance of the spatial distribution, the agent completes its final epoch rather than truncating it mid-streak. Because the maximum length of an epoch is D_j , the total step budget overshoots by at most D_j , remaining $\Theta(D_j^2)$.
- (iv) If the target x^* is not found by the end of the step budget, the agent increments j to $j + 1$ and begins the next phase from its current location.

We seek to bound $\mathbb{E}[C_{x^*}]$, the expected total number of single-cell steps taken across all phases before the target x^* is visited.

“Activation Phase” and Spectral Gap

In early phases where $D_j \ll D$, the agent’s streaks are too short to efficiently mix across the longest dimension of the torus. However, because D_j doubles every phase, there will be a specific phase j^* , which we call the *Activation Phase*, that is the first phase to satisfy $D_{j^*} \geq D$. By definition of the doubling sequence, $D \leq D_{j^*} < 2D$.

During Phase j^* , the sequence of epoch *start points* forms a discrete-time Markov chain on the abelian group S , where each transition corresponds to a full epoch trajectory. The transition kernel is translation-invariant. Because the state space is an abelian group, the

transition matrix P is a group convolution operator and is therefore normal ($PP^* = P^*P$). While not necessarily symmetric, its normality ensures the uniform distribution $\pi(x) = 1/|S|$ is its unique stationary measure, and its spectral properties mirror the strong mixing bounds of reversible chains.

The eigenvalues of this kernel correspond to the Fourier characters of $\mathbb{Z}_N \times \mathbb{Z}_M$. For any frequency vector $w = (w_1, w_2) \neq (0, 0) \in S$, the associated eigenvalue is:

$$\lambda_w = \frac{1}{4} \sum_{v \in V} \left(\frac{1}{D_{j^*}} \sum_{\ell=1}^{D_{j^*}} \exp \left(2\pi i \ell \left(\frac{v_1 w_1}{N} + \frac{v_2 w_2}{M} \right) \right) \right)$$

Let

$$\theta_v(w) := \frac{v_1 w_1}{N} + \frac{v_2 w_2}{M}.$$

Because V contains both axial and diagonal vectors, any non-zero frequency w guarantees a phase shift bounded away from 0 along at least one direction. For this non-aligned vector, the magnitude of the inner geometric sum is bounded by:

$$\frac{1}{D_{j^*}} \left| \frac{\sin(\pi D_{j^*} \theta)}{\sin(\pi \theta)} \right|$$

Because the minimum non-zero phase θ is $\Omega(1/M) = \Omega(1/D)$, and $D_{j^*} \geq D$, the summation covers enough of the group to destructively interfere. The magnitude of this sum is strictly bounded above by $1 - \delta$ for some constant $\delta > 0$ depending only on R .

Therefore, the eigenvalues are bounded strictly away from the unit circle: $|\lambda| \leq 1 - \frac{\delta}{4}$. This constant spectral gap ensures that the chain exhibits strong mixing properties comparable to reversible chains, with an effective relaxation time of $\mathcal{O}(1)$ epochs.

Stationary Hitting Probability

During Phase j^* , the expected length of an epoch is

$$\mathbb{E}[L] = \frac{D_{j^*} + 1}{2}.$$

Because $D_{j^*} < 2D = 2M$, a streak along the longest dimension wraps at most once. However, along the shorter dimension N , it may wrap up to $\lfloor 2M/N \rfloor \leq 2R$ times. Since the aspect ratio R is bounded by a constant, the number of distinct nodes explored in an epoch is reduced by at most a factor of $2R$. Thus, the number of distinct nodes remains strictly proportional to its length in an asymptotic sense.

If a trajectory begins from a start point drawn from the stationary uniform distribution π , the stationary probability $P_\pi(H)$ that this trajectory intersects x^* is bounded below by accounting for the maximum overlap:

$$P_\pi(H) \geq \frac{1}{2R} \sum_{\ell=1}^{D_{j^*}} P(L = \ell) \frac{\ell}{|S|} = \Theta \left(\frac{D_{j^*}}{|S|} \right)$$

For a normal Markov chain on a vertex-transitive graph, the expected number of epochs to hit x^* from stationarity is bounded by

$$\mathcal{O} \left(\frac{1}{1 - \max |\lambda|} + \frac{1}{P_\pi(H)} \right).$$

Given the constant spectral gap, this simplifies to:

$$\mathbb{E}_\pi[T_{x^*}] \leq \mathcal{O} \left(1 + \frac{|S|}{D_{j^*}} \right) = \mathcal{O} \left(\frac{|S|}{D_{j^*}} \right)$$

Phase j^* allocates a total step budget of $\Theta(D_{j^*}^2)$. Given the expected epoch length $\Theta(D_{j^*})$, the phase executes $\Theta(D_{j^*})$ epochs. Because $D_{j^*} \geq D$, and the bounded aspect ratio ensures $D^2 = \Omega(|S|)$, we have:

$$\text{Allocated Epochs} = \Theta(D_{j^*}) \geq \Omega\left(\frac{|S|}{D_{j^*}}\right)$$

Thus, the number of epochs allocated in Phase j^* meets or exceeds the expected number of epochs required to hit the target. Consequently, the probability p of finding the target during Phase j^* is bounded strictly away from zero by a constant (i.e., $p > 0$).

1 Bounding the Expected Total Steps

If the agent fails to find the target in Phase j^* , it proceeds to Phase $j^* + 1$, where the allocated steps quadruple. Crucially, the hitting probability does not remain constant; the failure probability decays exponentially.

In Phase $j^* + k$, the allocated maximum dimension is $D_{j^*+k} = 2^k D_{j^*}$, and the step budget is $\Theta(4^k D_{j^*}^2)$. The expected length of an epoch scales to $\Theta(2^k D_{j^*})$, meaning the phase allocates $K = \Theta(2^k D_{j^*})$ epochs.

As the expected epoch length exceeds the maximum dimension of the torus, the hitting probability per epoch saturates. A single epoch of infinite length along a random generator covers either a full row or a full column, capping the expected epochs required to hit the target from stationarity at $\mathbb{E}_\pi[T_{x^*}] \leq \mathcal{O}(M) = \mathcal{O}(|S|/D_{j^*})$.

Because the epoch start points form a normal Markov chain, we bound the non-hitting probability q_k using hitting time tail bounds for normal chains. The probability of failing to hit the target across K consecutive epochs decays exponentially relative to the expected hitting time:

$$q_k \leq \exp\left(-\Omega\left(\frac{K}{\mathbb{E}_\pi[T_{x^*}]}\right)\right)$$

Substituting $K = \Theta(2^k D_{j^*})$ and $\mathbb{E}_\pi[T_{x^*}] \leq \mathcal{O}(|S|/D_{j^*})$, we get:

$$q_k \leq \exp\left(-\Omega\left(\frac{2^k D_{j^*}^2}{|S|}\right)\right)$$

Because $D_{j^*}^2 = \Omega(|S|)$, the base fraction $\frac{D_{j^*}^2}{|S|} = \Omega(1)$. Thus, the failure probability decays exponentially:

$$q_k \leq \exp(-\beta 2^k)$$

for some constant $\beta > 0$.

The total number of single-cell steps taken up to the end of Phase j^* is dominated by a geometric series:

$$\sum_{i=1}^{j^*} \Theta(D_i^2) = \sum_{i=1}^{j^*} \Theta(4^i) = \Theta(4^{j^*}) = \Theta(D_{j^*}^2)$$

The expected total step cost is the sum of the maximum steps taken up to Phase j^* plus the expected cost of all subsequent phases:

$$\mathbb{E}[C_{x^*}] \leq \mathcal{O}(D_{j^*}^2) + \sum_{k=1}^{\infty} q_{k-1} \cdot \mathcal{O}(4^{j^*+k})$$

Substituting the bound on q_{k-1} :

$$\mathbb{E}[C_{x^*}] \leq \mathcal{O}(D_{j^*}^2) + \sum_{k=1}^{\infty} \exp(-\beta 2^{k-1}) \cdot \mathcal{O}(4^{j^*+k})$$

Because the exponential decay of $\exp(-\beta 2^{k-1})$ vastly overpowers the geometric growth of 4^k , the series converges easily to a constant multiplier of $\mathcal{O}(4^{j^*}) = \mathcal{O}(D_{j^*}^2)$.

$$\mathbb{E}[C_{x^*}] = \mathcal{O}(D_{j^*}^2) = \mathcal{O}(D^2) = \mathcal{O}(|S|)$$

This concludes the proof. The blind agent successfully achieves linear hitting time, adapting to the unknown bounds of the torus exclusively through continuous parameter scaling.